INCH-POUND

MIL-E-48441A(AR)
AMENDMENT 2
17 June 1991
SUPERSEDING
AMENDMENT 1
30 April 1979

MILITARY SPECIFICATION

ELBOW ASSEMBLY, PASSIVE: 11733401

This amendment forms a part of Military Specification MIL-E-48441A(MU), dated 31 December 1975, and is approved for use by the U.S. Army Armament, Munitions, and chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Delete and substitute:

"1.1 Scope. This specification covers a modular unit of the Tank Commander's M36E1 Periscope or the Tank Gunnerfs M32E1, M32CE1 and M35E1 Periscope, herein after referred to as Elbow Assembly (see 6.1)."

2.1 SPECIFICATIONS, Add:

"MIL-O-13830 Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of."

DRAWINGS, Delete:

"F11746721 Boot Objective
C11746729 Holder, Eyepiece 10" Collimator
D11746730 Eyepiece Assembly 10" Collimator".

PAGE 3

3.6.1.2, line 2, Delete "and 3.6.1.2.2".

PAGE 4

3.6.1.2.1, Delete and substitute:

"3.6.1.2.1 On axis. - The objective system shall resolve 40, 150 and 300 line pairs per millimeter on axis."

3.6.1.2.2, Delete in its entirety.

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3.6.5.1, Delete and substitute:

"3.6.5.1 Vibration "A". - This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to the contact for the requlator and the reticle projector lamp (Reference Drawing F11733401). Vibration shall be applied along each of three mutually perpendicular axes of the elbow assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .5G ± 10%.

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .5G to 1.5 G's ± 10%.

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's ± 10%.

d. During frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's ± 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .5G ± 10%.

f. At 5 Hz, the acceleration shall be .5G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the elbow assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive and 3.6.6 through 3.8 inclusive."

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3.6.5.2, Delete and substitute:

"3.6.5.2 Vibration "B". - This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401). Vibration shall be applied along each of three mutually perpendicular axes of the elbow assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain Constant acceleration of 1.0 G ± 10%.

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c. During a frequency sweep of 500 to 5 Hz, maintain COnstant acceleration of 1.0 G ± 10%.

d. At 5 Hz, the acceleration shall be 1.0 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes) . There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the elbow assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive and 3.6.6 through 3.8 inclusive."

3.6.5.3, Delete and substitute:

"3.6.5.3 Shock. - This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawig F11733401. The elbow assembly shall be subjected to 24 cumulative shocks consisting of three each of 35 G's ± 10%, then 20 G's ± 10% applied in each direction along each of the two mutually perpendicular horizontal axes (all for sides). The shock pulse shall have a half sine wave with a time duration of 3 ± .2 milliseconds. The elbow assembly shall also be subjected to six cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in one direction along the verticle axis (normal operating position of elbow assembly only). The shock pulse shall have a half sine wave with a time duration of 18 ± 3 milliseconds. Subsequent to shock, there shall be no failure or flicker of light emission from the image tube nor the reticle projector lamp. The elbow assembly shall also show no evidence of physical failure. Following the shock test the elbow assembly shall meet the requirements of 3.2, 3.6 through 3.6.4.10 inclusive an 3.6.6 through 3.8 inclusive.

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* 3.8, Delete and substitute:

"3.8 Sealing. - The elbow assembly shall show no evidence of leakage in excess of 0.1 pound per square inch gage (psig) when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 1 hour. The interior atmosphere of the accepted elbow assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig)."

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4.3.2.1.1, lines 2 and 3 Delete “image intensifier tube.”

TABLE I, Delete characteristic “302” in its entirety.

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4.5.2.3.2, Delete and substitute:

“4.5.2.3.2 Performance. - Special test equipment to determine performance of the elbow assembly shall conform to the requirements Of Drawing F11746736 (Test Set-up for M35 Elbow) , Drawing F11746724 (Holder, eyepiece) and drawings pertaining thereto.”

4.6.2.1, Delete and substitute:

“4.6.2.1 Resolution-objective system-on axis. - Perform this test utilizing a resolving power chart on a wall target and an auxiliary telescope having a magnification of at least 50 power (50X) and a numerical aperture of at least .65. The resolving power chart shall represent the angular subtense for line pairs per millimeter specified in 3.6.1.2.1 and shall contain four line sets as shown in Figure I of MIL-O-13830. The test method and procedure shall be in compliance with the “Resolution Test” specified in MIL-O-13830. The limits of resolution shall be within the value specified in 3.6.1.2.1, as determined by viewing through the auxiliary telescope.”

PAGE 15

4.6.2.2. Delete in its entirety.

PAGE 18

4.6.5.2, line 21, Delete “M32E1, M35E1 and M36E1” and substitute “M32E1, M32CE1, M35E1 and M36E1.”

PAGE 19

4.6.5.5.1, line 21, Delete “M32E1, M35E1 and M36E1” and substitute “M32E1, M32CE1, M35E1 and M36E1”

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4.6.6.5.2, line 12, Delete “M32E1, M35E1, and M36E1” and substitute “M32E1, M32CE1, M35E1 and M36E1”

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* 4.6.8, line 2, Delete “0.25 to 0.50” and substitute “0.0 to 0.50”

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6.1 line 3, Delete Periscope: M32E1 and M35E1" and substitute “Periscope: M32E1, M32CE1 and M35E1"

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